

Response Letter with Annotations

Mukerjee, S., Yang, T., & Peng, Y. (2023). Metrics in action: How social media metrics shape news production on Facebook. *Journal of Communication*, 73(3), 260-272.

<https://doi.org/10.1093/joc/jqad012>

Annotations

1. Purposes

a. Motivation (one-sentence summary)

- The key motivation of this study is to empirically investigate whether and how real-time social media engagement metrics (such as likes and shares) function as a feedback loop that pressures news organizations to alter their future agendas to favor high-performing topics.

b. Specific Rationale (several bullet points)

- Engagement metrics on social media has affected news industries and journalism significantly: shaping journalistic cultures, values and norms; shaping journalistic labor processes; shaping journalistic practices.
 1. However, under the competing frameworks of the Trustee Model and the Market Model, the specific ways in which newsrooms respond to these metrics, and whether they fundamentally alter the news supply, remain understudied.
- A key rationale is exploring whether different political media ecosystems respond to metrics differently. The authors hypothesize that right-wing outlets might be more reactive to audience feedback as a way to build "identity-based" connections and bypass traditional mainstream journalistic standards.
- There is a normative concern that if metrics drive production, newsrooms will shift resources toward entertainment-oriented "soft news" or highly divisive partisan topics to boost revenue. The study motivates the need to understand how this process might inadvertently marginalize important public-interest reporting.

c. Research Questions and/or Hypotheses

- Identify IV, DV, mediator, moderator
 1. IV
 - a. Engagement Signal
 2. DV
 - a. News Content Production (Frequency)
 3. Moderator
 - a. Topic Type
 - b. Ideological Slant (Media Partisanship)
- Hypotheses
 1. **H1:** Engagement metrics on social media affect news content production. News topics that attract greater engagement are increasingly covered in subsequent time periods on Facebook.

2. **H2:** Media outlets are more likely to respond to audience engagement with partisan politics topics (H2a) and entertainment-oriented topics (H2b) compared to other topics.
3. **H3:** Responsiveness of outlets to topics will be associated with their ideological slants: right wing outlets are more likely to respond to audience metrics than left-wing outlets.
4. **H4:** Responsiveness of outlets to audience engagement with partisan topics will be associated with their ideological slants: right-wing outlets are more likely to respond to audience metrics received by partisan news posts than leftwing outlets.
5. **RQ1:** How is the responsiveness of an outlet to audience engagement with entertainment-oriented topics related to its ideological slant?

2. Method

a. Research Design

- The study employs a large-scale longitudinal design to examine the dynamic relationship between audience engagement and news production.
- Data Source: All Facebook posts published by 29 major U.S. media outlets over a five-year period (January 2015 – December 2019).
- Dataset Size: Approximately 2.23 million posts, retrieved via CrowdTangle.
- Unit of Analysis: The researchers divided the five-year period into 60 one-month windows to identify evolving news topics.

b. Measures

- Independent Variable: Engagement Signal
 1. Defined as the sum of comments, shares, and reactions (likes, love, wow, etc.) for a specific topic at time. It is calculated relative to the average engagement of all other posts in that period to identify what was "trending".
- Dependent Variable: Topic Frequency
 1. The volume or count of posts published by an outlet for a specific topic on the subsequent day.
- Moderators (Classification):
 1. Topic Type: Topics were manually coded as Partisan Politics (e.g., abortion, gun control), Entertainment-oriented (e.g., celebrities, life hacks), or Other.
 2. Media Bias: Outlets were assigned an ideology score (1 to 5) based on Pew Research Center survey data regarding the ideological composition of their audience.

c. Analysis

- Topic Modeling (NMF):
 1. The researchers used Non-Negative Matrix Factorization (NMF) to identify 1,005 unique topics across the 60 months.

2. NMF was chosen because it is particularly effective for short social media texts.
- Statistical Modeling (Fixed-Effects Regression):
 1. The core analysis used a Fixed-Effects Model with a lagged dependent variable.
 2. Controls: The model controlled for "news value" (how much the same topic was being covered by other outlets) and included fixed effects for the outlet, time, and topic to isolate the specific impact of metrics.
 3. Time Lag: The primary analysis used a 1-day lag, meaning it tested if high engagement today led to more stories tomorrow

3. Results and Conclusions

a. Answers to RQs and Hypotheses

- H1: **Supported.** The engagement signal received for a topic at time $t-1$ had a significant positive effect on its coverage at time t .
- H2a: **Not Supported.** Interaction effects showed that topic type (partisan vs. other) did not moderate the *overall* responsiveness across the entire media landscape.
- H2b: **Not Supported.** Similar to H2a, there was no overall significant moderating effect for entertainment topics in the primary analysis.
- H3: **Supported.** A significant correlation ($r = 0.412$) was found between responsiveness and ideological slant, driven primarily by a few highly responsive right-wing outlets like Breitbart and Washington Examiner.
- H4: **Supported.** There was a significant correlation ($r = 0.477$) between media slant and responsiveness specifically for partisan topics.

b. Include Key Figures/Tables

Table 1. OLS regression estimates under various model specifications
Clustered standard errors in parenthesis.

	DV: Frequency of topic at time t in media m			
	Model 1	Model 2	Model 3	Model 4
ES ^a with topic for media m at $t - 1$	0.0022** (0.0006)	0.0019** (0.0006)	0.0019** (0.0006)	0.0017** (0.0006)
Covariates				
Frequency of topic at $t - 1$ in media m	0.6563*** (0.0309)	0.6555*** (0.0313)	0.6658*** (0.0365)	0.6657*** (0.0365)
Avg. ES with topic for all outlets		0.0053 (0.0039)	0.0098*** (0.0024)	0.0098*** (0.0024)
Freq. of topic across all outlets at $t - 1$			-0.0666 (0.0404)	-0.0665 (0.0403)
Fixed effects for media	Yes	Yes	Yes	Yes
Fixed effects for time	Yes	Yes	Yes	Yes
Fixed effects for topic	Yes	Yes	Yes	Yes
Interactions				
ES with topic for media m at $t - 1 \times$ partisan topic				0.0000 (0.0005)
ES with topic for media m at $t - 1 \times$ entertainment topic				0.0009 (0.0007)

^a Engagement Signal (log transformed) as defined in the text.
⁺ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

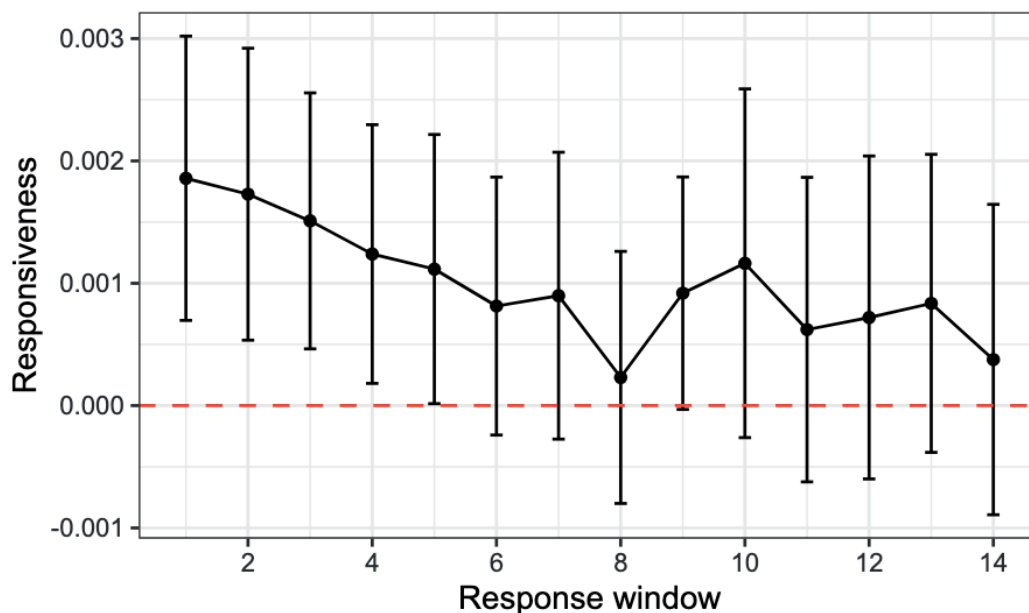


Figure 1. Point estimates for responsiveness scores (and 95% confidence intervals) for response windows ranging from 1 day to 14 days.

Note: Media outlets are, on average, more responsive to engagement signals picked up over short periods of time, than to those aggregated over several days.

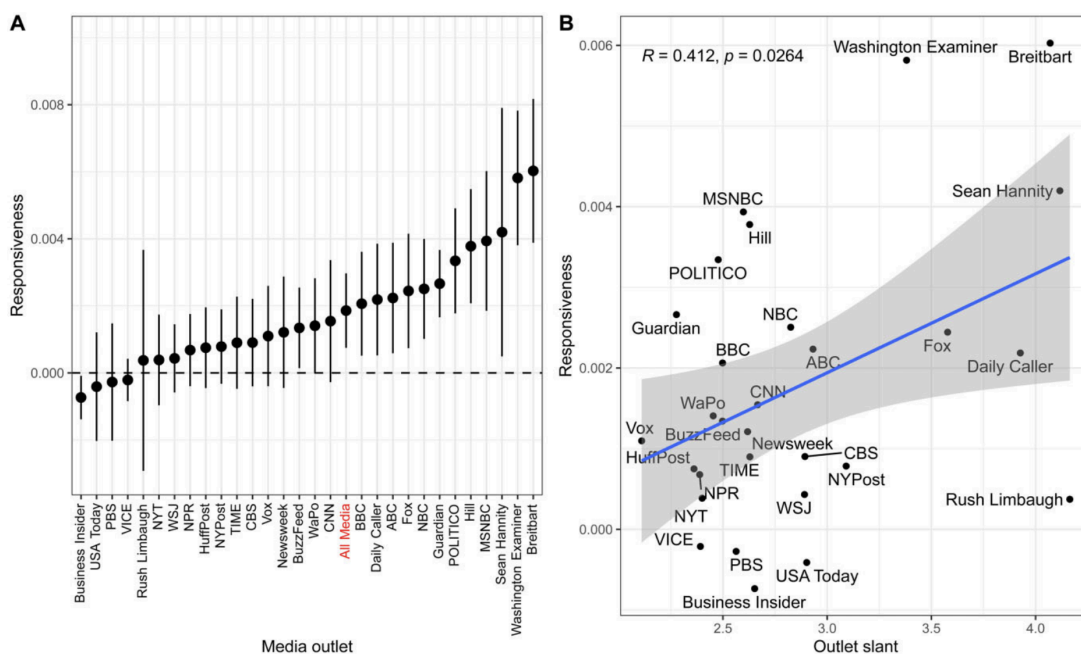


Figure 2. Responsiveness scores for individual outlets (A) and scatterplot showing correlation between responsiveness and ideological slant scores (B).
Note: Conservative outlets are significantly more likely to respond to audience engagement signals than liberal outlets.

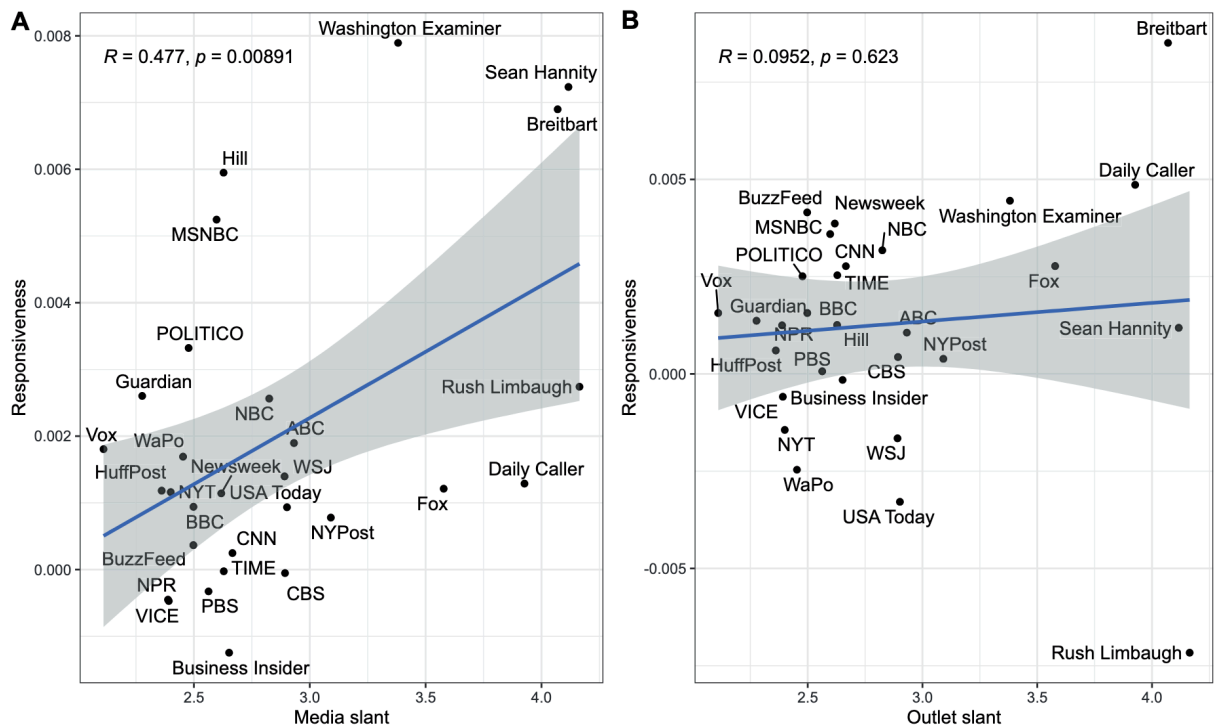


Figure 3. Correlation between ideological slant scores and responsiveness values to partisan topics (A) and entertainment-oriented topics (B).
 Note: Conservative outlets are significantly more likely to respond to audience engagement signals than liberal outlets, but only to partisan topics.

c. Authors' Conclusions

- The study reveals that social media engagement metrics create a significant but modest "feedback loop" that shapes news production, where high engagement for a topic today leads to increased coverage of that topic tomorrow. However, this effect is highly asymmetrical; while most legacy and mainstream media remain relatively resistant to "click-chasing," a handful of right-wing outlets (e.g., Breitbart, Washington Examiner) are hyper-responsive to metrics, particularly when it involves partisan politics. Interestingly, when it comes to entertainment-oriented "soft news," all media outlets, regardless of their ideological slant, show a similar tendency to follow the metrics.

4. Discussion

a. Implications

- The authors emphasize that while metrics do influence news, they haven't completely taken over. Traditional news values still act as a buffer.
- A major takeaway is that right-wing media are much more sensitive to clicks. They use metrics as a "market signal" to double down on partisan topics that fire up their audience.

- For non-political "soft news" (like celebrities), everyone acts the same. In this area, the "Market Model" wins across the board, regardless of an outlet's politics.

b. Limitations

- The findings are limited to Facebook; platforms like TikTok or X (Twitter) might have different dynamics.
- The data shows what happened, but it can't see inside the newsroom to know why editors made those specific choices.
- The study treats all engagement (likes, angries, shares) as one signal, but different reactions might drive different editorial responses.

c. Future Directions

- Future research should compare how different social media platforms pressure journalists in different ways.
- Researchers should look at specific emotions to see if "outrage" drives more news production than "positivity."
- The authors suggest combining big data with interviews to understand the human side of how metrics are used in daily meetings.

d. ...

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Response Letter (~600 words)

- Your assessment of the study
 - I think this study is very interesting because it uses a strong theory to explain a modern problem. It connects the Trustee Model (journalists do what is right) and the Market Model (journalists do what sells) to show how newsrooms are changing. I am very impressed by the “big data” they used. They collected 2.23 million Facebook posts from 29 major U.S. media outlets over five years. This is a huge amount of work!
 - The most amazing part for me is the method. Usually, topic modeling is just one big group of data, but this paper is different. The authors split the five-year period into 60 monthly windows. They used a method called NMF (Non-negative Matrix Factorization) instead of the common LDA. I think this is very smart because news changes every month. What was popular in 2015 is not the same as in 2019. By doing it month by month, they found 1,005 specific news topics. This helps the study stay very close to the real news context.
 - Also, I like how they checked their work. They did not just trust the computer. They used human coding to check 200 topics to make sure the machine was right. Their findings about ideological asymmetry are very important. It shows that right-wing outlets like Breitbart and Washington Examiner are hyper-responsive to metrics. They chase clicks much more than legacy media like The New York Times. This proves that social media metrics are reshaping the news agenda, but the effect depends on the political slant of the newsroom.
- Weaknesses
 - Even though this is a great paper, there are some weaknesses. First, the study only uses data from Facebook. We know that news on Twitter, TikTok, or Instagram is very different. Users on those platforms have different behaviors, so we cannot say the “feedback loop” works the same way everywhere.
 - Second, the way they measure “engagement” is a bit simple. They added likes, shares, and comments together into one “Engagement Signal.” But in real life, a “share” is much more powerful than a “like.” Also, Facebook has different reactions like “Haha,” “Sad,” or “Angry.” An “Angry” reaction might make a newsroom produce more negative or polarized news, but this study doesn't look at the specific emotions. It treats all clicks as the same thing.
 - Third, the study only looks at the data from the outside. The authors call this “trace data.” It tells us what happened, but it doesn't tell us why. We don't know if the journalists were looking at the screens and feeling stressed, or if their bosses forced them to write more viral stories. We can't see the internal meetings or the “human” side of the newsroom.
 - Finally, the 1-day lag they used for analysis is very short. While the study shows that today's clicks affect tomorrow's news, it doesn't tell us about the long-term effects. Does chasing clicks for years make the quality of journalism lower overall? The paper doesn't answer if this leads to a race for all news organizations.
- Other notes

